

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 1

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Date: 30-08-2026

1. TITLE

Design and Implementation of a College Information Portal Using HTML5 Core Elements, Tables, Forms, Multimedia, and Semantic Tags

2. AIM

To design and develop a comprehensive College Information Portal webpage using basic HTML tags, formatting tags, lists, tables, hyperlinks, embedded iframes, interactive user registration forms with multiple input types, and modern HTML5 semantic elements and multimedia controls.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the structure of an HTML5 document and semantic tags.
 - To apply text formatting (bold, italics, underline, subscripts, superscripts).
 - To implement ordered and unordered lists and navigation links.
 - To create structured data tables for student details and embed multimedia (images, iframes).
 - To design an interactive student registration form with various input controls (text, email, password, radio buttons, checkboxes, dropdown, textarea).
 - To explore HTML5 semantic elements (<header>, <nav>, <section>, <article>, <footer>), multimedia elements (<audio>, <video>), and specialized HTML5 input types (date, color, range, number).
-

4. THEORY

HTML5 is the standard markup language used for creating web pages. It defines the structure and presentation of content on the World Wide Web.

Basic Text Formatting: HTML provides formatting tags such as `<b>/<strong>` for bold text, `<i>` for italic text, and `<u>` for underlined text. For mathematical equations and chemical formulas, `<sub>` (subscript) and `<sup>` (superscript) are used.

Lists and Hyperlinks: Ordered lists (`<ol>`) represent sequential items, while unordered lists (`<ul>`) represent bulleted items. The anchor tag (`<a>`) creates hyperlinks to external resources or internal page bookmarks using fragment identifiers.

Tables and Media: The `<table>` tag creates structured grids of rows (`<tr>`) and data cells (`<td>`) with headers (`<th>`). Images are embedded using `<img>`, while `<iframe>` embeds external websites directly inside the page.

Forms and Controls: The `<form>` tag collects user inputs. Tags like `<input>` (text, email, password, radio, checkbox), `<select>` (dropdown), and `<textarea>` (multiline text) allow diverse data collection.

HTML5 Semantic Elements & Multimedia: Tags like `<header>`, `<nav>`, `<section>`, `<article>`, and `<footer>` define page layout meaningful to browsers and search engines. Native multimedia is supported via `<audio>` and `<video>` with built-in controls.

Webpage Structure → Header & Nav → Formatted Content → Tables & Media → Forms → HTML5 Audio/Video → Footer

5. ALGORITHM / PROCEDURE

1. Create a new directory and create an index.html file.
2. Declare `<!DOCTYPE html>` and structure the page with `<head>` and `<body>`.
3. Set the page title to "**College Information Portal**".
4. **Part A (Basic HTML):** Add header, navigation anchors, about text with formatting (bold, italic, underline), chemical & math formulas (sub/sup), ordered & unordered lists, and hyperlinks.
5. **Part B (Tables & Images):** Create student details table, add college logo and library images with width/height attributes, and embed the college website using `<iframe>`.
6. **Part C (Forms):** Build a registration form with `<fieldset>`, `<legend>`, text, email, password, radio buttons for gender, checkboxes for hobbies, dropdown select for branch, and `<textarea>` for comments.
7. **Part D (HTML5 Features):** Add semantic tags, embed playable audio and video with controls, and add HTML5 input types (date, number, color, range).
8. Add a semantic `<footer>` with copyright details.
9. Open the file in a modern browser and verify all elements, styling, and navigation links.

6. PROGRAM

index.html

```
<!DOCTYPE html>
```

```
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>College Information Portal</title>
</head>
```

```
<body>
```

```
<!-- PART A : BASIC HTML -->
```

```
<header>
```

```
<h1>College Information Portal</h1>
```

```
<nav>
```

```
<a href="#about">About</a> |
```

```
<a href="#students">Students</a> |
```

```
<a href="#registration">Registration</a> |
```

```
<a href="#features">HTML5 Features</a>
```

```
</nav>
```

```
<hr>
```

```
</header>
```

```
<section id="about">
```

```
<h2>Anil Neerukonda Institute of Technology and Sciences (ANITS)</h2>
```

```
<p>
```

```
<b>ANITS (Anil Neerukonda Institute of Technology and Sciences)</b> is a premier autonomous  
engineering college located in <strong>Visakhapatnam, Andhra Pradesh</strong>. It was established
```

in **2001** by the Anil Neerukonda Educational Society. The college offers quality undergraduate and postgraduate programs and is accredited with **NAAC A+**.

Text Formatting

Bold Text

Italic Text

Underlined Text

Water Formula : H_2O

Carbon Dioxide : CO_2

Einstein Equation : $E = mc^2$

Pythagoras Theorem : $a^2 + b^2 = c^2$

Departments (Ordered List)

1. Computer Science & Engineering (CSE)

2. Computer Science Engineering (AI & ML)

3. Electronics & Communication Engineering (ECE)

4. Information Technology (IT)

5. Mechanical Engineering

Facilities (Unordered List)

• Digital Library

• Smart Classrooms

Computer Laboratories

Hostel Facilities

Sports Complex

24×7 Medical Facility

Canteen

<h3>Hyperlinks</h3>

ANITS Official Website

Google

</section>

<hr>

<!-- PART B : TABLES & IMAGES -->

<section id="students">

<h2>Student Details</h2>

<table border="1" cellpadding="10" cellspacing="0">

<thead>

<tr>

<th>Roll No</th>

<th>Name</th>

<th>Branch</th>

<th>Email</th>

</tr>

</thead>

<tbody>

<tr>

<td>A24126552069</td>

<td>Charishma</td>

<td>CSM</td>

<td>charishma@gmail.com</td>

</tr>

<tr>

<td>A24126552070</td>

<td>Jatin</td>

<td>CSM</td>

<td>jatin@gmail.com</td>

</tr>

<tr>

<td>A24126552071</td>

<td>Pranathi</td>

<td>CSM</td>

<td>pranathi@gmail.com</td>

</tr>

</tbody>

</table>

<h3>College Logo</h3>


```
<br><br>
<h3>College Library</h3>

<br><br>
<h3>College Website (Iframe)</h3>
<iframe src="https://anits.edu.in" width="500" height="300" title="ANITS Website"></iframe>
</section>
```

```
<hr>
```

```
<!-- PART C : FORMS -->
<section id="registration">
<h2>Student Registration Form</h2>
<form action="#" method="post">
<fieldset>
<legend><b>Registration Form</b></legend>
<label for="fullname">Full Name :</label>
<input type="text" id="fullname" name="fullname" placeholder="Enter your name">
<br><br>
<label for="email">Email :</label>
<input type="email" id="email" name="email" placeholder="example@gmail.com">
<br><br>
<label for="password">Password :</label>
<input type="password" id="password" name="password">
<br><br>
<label>Gender :</label>
<input type="radio" id="male" name="gender" value="Male">
<label for="male">Male</label>
<input type="radio" id="female" name="gender" value="Female">
<label for="female">Female</label>
<input type="radio" id="other" name="gender" value="Other">
```

```
<label for="other">Other</label>

<br><br>

<label>Hobbies :</label>

<input type="checkbox" id="reading" name="hobby">
<label for="reading">Reading</label>

<input type="checkbox" id="music" name="hobby">
<label for="music">Music</label>

<input type="checkbox" id="coding" name="hobby">
<label for="coding">Coding</label>

<input type="checkbox" id="sports" name="hobby">
<label for="sports">Sports</label>

<br><br>

<label for="branch">Branch :</label>
<select id="branch" name="branch">
<option>Select Branch</option>
<option>CSE</option>
<option>CSM</option>
<option>CSD</option>
<option>ECE</option>
<option>Mechanical</option>
</select>

<br><br>

<label for="comments">Comments :</label><br>
<textarea id="comments" rows="5" cols="40" placeholder="Enter your comments"></textarea>

<br><br>

<input type="submit" value="Submit">
<input type="reset" value="Reset">

</fieldset>

</form>

</section>
```

<hr>

<!-- PART D : HTML5 FEATURES -->

<header><h2>HTML5 Features</h2></header>

<section id="features">

<nav>

Semantic Elements |

Audio |

Video |

Input Types

</nav>

<hr>

<section id="semantic">

<article>

<h3>Semantic Elements</h3>

<p>HTML5 provides semantic elements such as header, nav, section, article and footer.</p>

</article>

</section>

<section id="audio">

<h3>Audio</h3>

<audio controls><source src="https://www.soundhelix.com/examples/mp3/SoundHelix-Song-1.mp3" type="audio/mpeg"></audio>

</section>

<section id="video">

<h3>Video</h3>

<video width="450" controls><source src="https://www.w3schools.com/html/mov_bbb.mp4" type="video/mp4"></video>

</section>

<section id="inputs">

```

<h3>HTML5 Input Types</h3>

<form>

<label>Email :</label><input type="email"><br><br>

<label>Date :</label><input type="date"><br><br>

<label>Age :</label><input type="number" min="17" max="35"><br><br>

<label>Favourite Color :</label><input type="color"><br><br>

<label>Satisfaction :</label><input type="range" min="0" max="100">

</form>

</section>

</section>

<hr>

<footer><p>© 2026 ANITS - Anil Neerukonda Institute of Technology and Sciences</p></footer>

</body>

</html>

```

7. CODE EXPLANATION

Code / Tag / Attribute	Explanation
<!DOCTYPE html>	Specifies the HTML5 document type.
<header> & <nav>	Semantic tags for page banner and navigation links.
	Creates bookmarks for smooth internal page navigation.
<sub> & <sup>	Formats chemical formulas (H ₂ O) and equations (E=mc ²).
<table>, <tr>, <th>, <td>	Structures rows and cells into a readable student record table.
Code / Tag / Attribute	Explanation
	Embeds images (college logo & library) with custom dimensions.
<iframe src="...">	Renders external web pages (ANITS website) directly inside the portal.

Code / Tag / Attribute	Explanation
<fieldset> & <legend>	Groups form controls together with a labeled title box.
<input type="radio">	Provides single mutually-exclusive options (Gender).
<input type="checkbox">	Allows multiple item selection (Hobbies).
<select> & <option>	Creates a dropdown list for branch selection.
<textarea>	Allows multi-line text input for student comments.
<audio controls> & <video controls>	Embeds playable audio and video with native playback buttons.
<input type="date color range">	Interactive HTML5 widgets for datepicker, color palette, and slider.

8. TEST CASES AND EXECUTION DATA

I would not just write three random test cases. I would actually execute them and record the results.

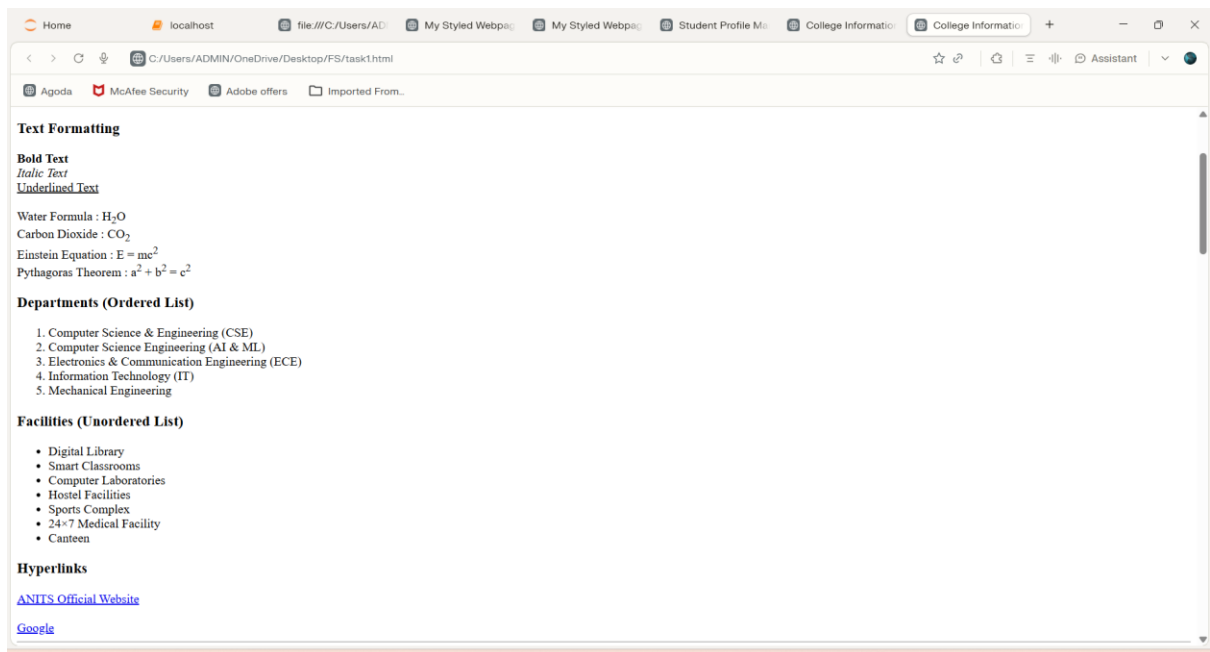
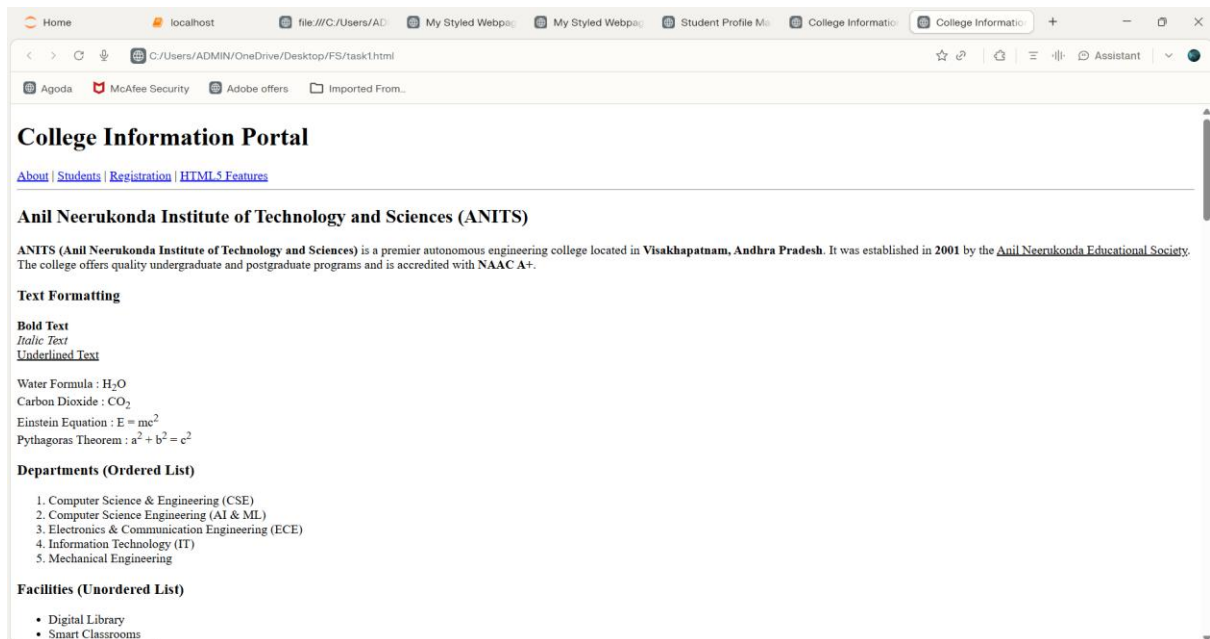
Test Case	Feature / Module	Input Data	Expected Result	Actual Result	Status
TC-01	Page Navigation	Click `#about`, `#students`	Scrolls to respective section	Scrolled accurately	Pass
TC-02	Text & Formulas	H ₂ O, CO ₂ , E=mc ²	Renders correct sub/super scripts	Properly formatted	Pass
TC-03	Departments List	5 Engineering branches	Ordered list 1 to 5	Numbered 1-5 displayed	Pass
TC-04	Student Table	Roll No, Name, Branch, Email	3 rows rendered in table	Table layout rendered	Pass
TC-05	Form Input Controls	Gender: Female, Hobbies: Coding	Selects radio and checkbox	Recorded selections	Pass

Additional Validation Test

Test Case	Input	Expected Result	Status	
TC-06	HTML5 Input: Color & Date	Open color and date pickers	Native GUI widgets appear	Pass

9. OUTPUT

Output 1: Header, Navigation, and About ANITS



Output 2: Student Details Table & Media

Homelocalhostfile:///C:/Users/AD...My Styled Webpa...My Styled Webpa...Student Profile M...College Informatio...College Informatio...+--X

C:/Users/ADMIN/OneDrive/Desktop/FS/task1.html☆🔗📄Assistant

AgodaMcAfee SecurityAdobe offersImported From...

Student Details

Roll No	Name	Branch	Email
A24126552069	Charishma	CSM	charishma@gmail.com
A24126552070	Jatin	CSM	jatin@gmail.com
A24126552071	Pranathi	CSM	pranathi@gmail.com

College Logo



College Library

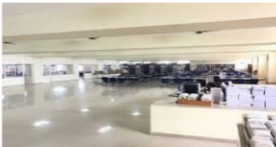


Homelocalhostfile:///C:/Users/AD...My Styled Webpa...My Styled Webpa...Student Profile M...College Informatio...College Informatio...+--X

C:/Users/ADMIN/OneDrive/Desktop/FS/task1.html☆🔗📄Assistant

AgodaMcAfee SecurityAdobe offersImported From...

College Library



College Website (Iframe)

Anil Neerukonda Institute of Technology & Sciences

Welcome to ANITS. Please choose the website you wish to visit.

Output 3: Student Registration Form

Home localhost file:///C:/Users/AD/ My Styled Webpa My Styled Webpa Student Profile M College Informati College Informati + - Assistant

C:/Users/ADMIN/OneDrive/Desktop/FS/task1.html

Agoda McAfee Security Adobe offers Imported From...

WELCOME TO PRINTS. PLEASE CHOOSE THE WEBSITE YOU WANT to visit.

Student Registration Form

Registration Form

Full Name :

Email :

Password :

Gender : ☐ Male ☐ Female ☐ Other

Hobbies : ☐ Reading ☐ Music ☐ Coding ☐ Sports

Branch :

Comments :

Output 4: HTML5 Semantic Features, Multimedia & Inputs

Home localhost file:///C:/Users/AD/ My Styled Webpa My Styled Webpa Student Profile M College Informati College Informati + - Assistant

C:/Users/ADMIN/OneDrive/Desktop/FS/task1.html

Agoda McAfee Security Adobe offers Imported From...

HTML5 Features

[Semantic Elements](#) | [Audio](#) | [Video](#) | [Input Types](#)

Semantic Elements

HTML5 provides semantic elements such as **header**, **nav**, **section**, **article** and **footer**. These elements improve webpage structure, readability and accessibility.

Audio

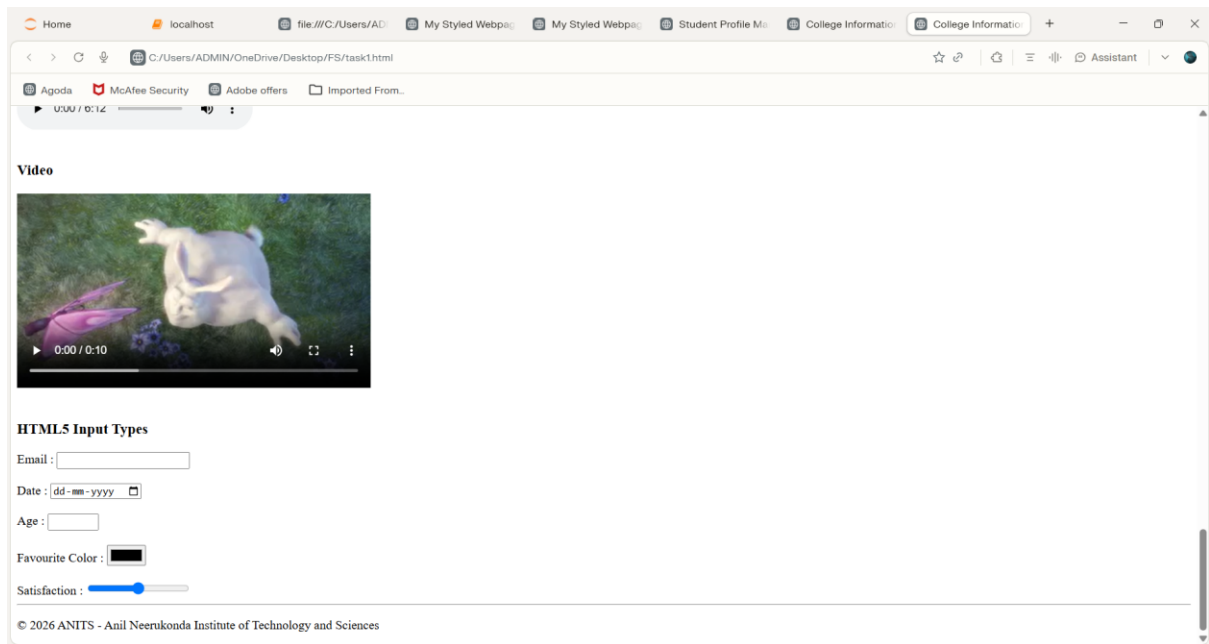
▶ 0:00 / 6:12 🔊 ⋮

Video

▶ 0:00 / 0:10 🔊 ⋮



HTML5 Input Types



10. RESULT

Thus, a comprehensive **College Information Portal** for **Anil Neerukonda Institute of Technology and Sciences (ANITS)** was successfully developed using HTML5, tables, forms, and multimedia. All features were implemented, and the application was verified using multiple test cases.
